Tanaka Certificates: Weekly Research Report

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Abstract

This week I investigated whether nonsmooth neural networks can serve as continuous-time supermartingale certificates. The main theoretical outcome is an Itô–Tanaka characterization for one-dimensional piecewise-smooth certificates and a failure result showing that piecewise-linear certificates cannot solve an important class of reach–avoid problems. Curvature removes this obstruction, which motivates either piecewise-quadratic certificates or a more simply, a network with a bunch of ReLU layers with a piecewise-quadratic in the end. I also implemented the first verifier scaffold for one-dimensional ReLU certificates and tested it on an Ornstein–Uhlenbeck and a simple Brownian motion example.

1 Problem setting

Consider a one-dimensional stochastic differential equation

$$dX_t = f(X_t)\,dt + g(X_t)\,dW_t \tag{1}$$

on a domain D. For a reach–avoid problem, let $X_0 \subseteq D$ be the initial set, $X_u \subseteq D$ the unsafe set, and $X_\tau \subseteq D$ the target. We seek a certificate V and constants $\alpha < \beta$ and $\varepsilon > 0$ such that

1. boundary condition holds:
$$\sup_{x\in X_0} V(x) \le \alpha \qquad \text{and} \qquad \inf_{x\in X_u} V(x) \ge \beta \tag{2}$$
2. generator condition holds:
$$\mathcal{L}V(x) \le -\varepsilon \qquad \text{for } x \in \{V \le \beta\} \setminus X_\tau, \tag{3}$$

where, at smooth points,

$$\mathcal{L}V(x) = V'(x)f(x) + \frac{1}{2}V''(x)g(x)^2. \tag{4}$$

The process is stopped after reaching the target or unsafe set: once the reach–avoid property has been achieved or violated, its later trajectory is irrelevant.

In short, these conditions ensure that $V(X_t)$ is a supermartingale (i.e., a decreasing process) that starts below α and is guaranteed to exceed β if the process reaches the unsafe set.

Previously we would assume $V \in \mathcal{C}^2$. ReLU networks are instead continuous and piecewise linear. Their second derivative vanishes inside each linear region and is singular at a kink, so the usual Itô formula does not directly capture their behaviour. This is the reason for studying Tanaka certificates.

2 Goal of the project

The first goal of this project is to determine whether nonsmooth verifiers (such as ReLU networks which potentially enable ultra-fast verification, or **even more interestingly: multiple certificates composed together using** max **and** min) can be used to solve reach–avoid problems.

Therefore my first task was very simple: analyze the one-dimensional case and determine whether piecewise-linear certificates are expressive enough to solve reach–avoid problems.

Within this scope, the verifier should check the two boundary conditions in (2), the decrease condition on the sub-β basin outside the target, and the additional conditions introduced at kinks.

3 Preliminary investigation

3.1 The Itô–Tanaka decomposition

Let V be continuous and piecewise $\mathcal{C}^2$, with finitely many breakpoints x_i. The Itô–Tanaka formula gives

$$\begin{aligned} V(X_t) = V(X_0) &+ \int_0^t \mathcal{L}V(X_s)\,ds + \int_0^t V'(X_s)g(X_s)\,dW_s \\ &+ \frac{1}{2}\sum_i \big(V'_+(x_i) - V'_-(x_i)\big) L_t^{x_i}(X), \end{aligned} \tag{5}$$

where $L_t^{x_i}(X)$ is local time. Subject to the usual integrability condition making the stochastic integral a martingale, two sign conditions are sufficient for $V(X_t)$ to be a supermartingale:

$$\mathcal{L}V(x) \le 0 \qquad \text{inside every smooth piece,} \tag{6}$$

$$V'_+(x_i) - V'_-(x_i) \le 0 \qquad \text{at every relevant kink.} \tag{7}$$

For a piecewise-linear V, condition (7) says that the slopes may only decrease: V must be concave on the region where the certificate condition is imposed; Figure 1 illustrates a valid and an invalid sequence of slopes.

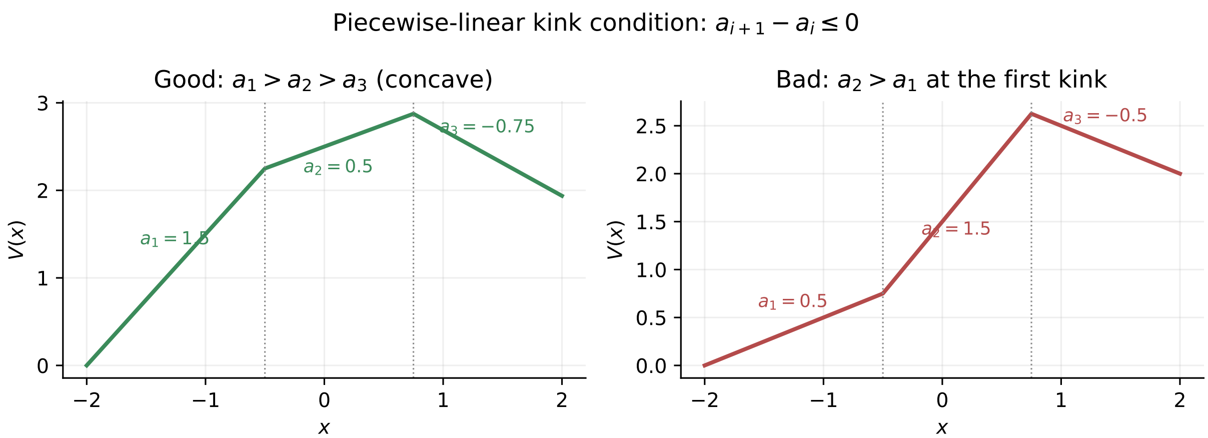

Figure 1: The Tanaka kink condition for a continuous piecewise-linear certificate. Decreasing slopes give nonpositive derivative jumps and hence a concave certificate (left). An increase in slope creates a positive derivative jump and violates the condition (right).

Meanwhile $V'' = 0$ inside every piece, so

$$\mathcal{L}V(x) = V'(x)f(x). \tag{8}$$

The fact that $V'' = 0$ inside every linear piece yields a problem, which is illustrated in the next subsection.

3.2 A motivating counterexample

For

$$dX_t = dt + \sqrt{2}\,dW_t, \qquad x \in [0,1], \tag{9}$$

suppose the boundary requirements are $V(0) = 0$ and $V(1) = 1$. Any continuous piecewise-linear function satisfying them must have positive slope somewhere. At such a point, (8) is positive, so the generator condition fails. In contrast, the normalized quadratic candidate

$$V_{\mathrm{quad}}(x) = 2x - x^2 \tag{10}$$

satisfies the same boundary conditions and

$$\mathcal{L}V_{\mathrm{quad}}(x) = V'_{\mathrm{quad}}(x) + V''_{\mathrm{quad}}(x) = -2x \le 0. \tag{11}$$

Thus the diffusion term can use negative curvature to compensate for positive drift, while a linear piece cannot. Figure 2 compares a concave piecewise-linear candidate, the quadratic candidate, and the exact martingale certificate under identical boundary conditions.

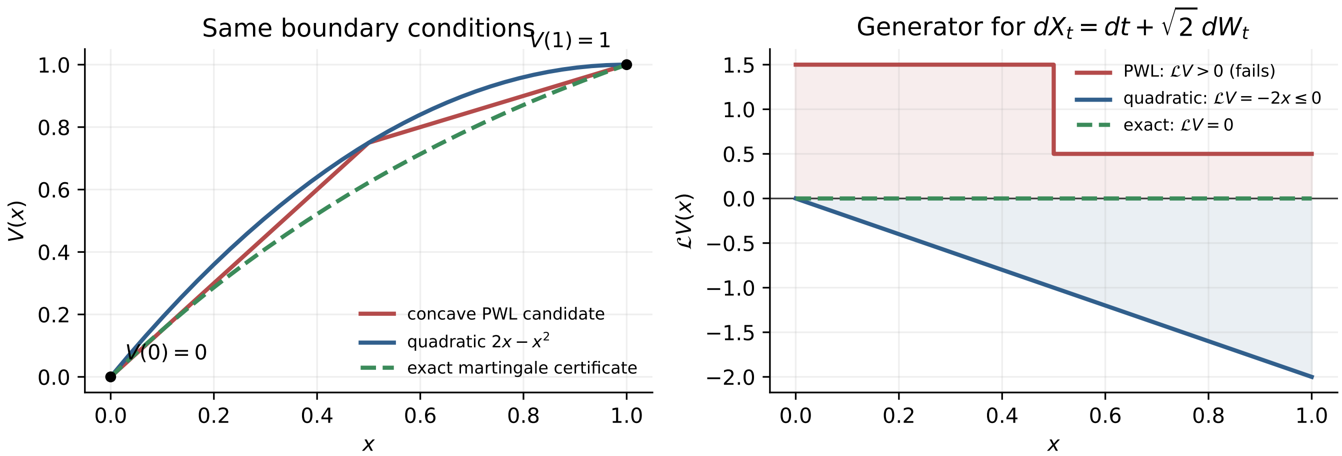

Figure 2: The motivating counterexample with all candidates normalized by $V(0) = 0$ and $V(1) = 1$. Even a concave piecewise-linear candidate has a positive generator on each affine piece. Negative curvature makes the quadratic generator nonpositive, while the exact certificate has zero generator.

The exact martingale certificate for this example solves $V' + V'' = 0$. With $V(0) = 0$ and $V(1) = 1$, it is

$$V_\star(x) = \frac{1 - e^{-x}}{1 - e^{-1}}. \tag{12}$$

Since

$$1 - e^{-x} = x - \frac{x^2}{2} + \frac{x^3}{6} - \cdots, \tag{13}$$

the normalized quadratic candidate is the correspondingly normalized second-order Taylor approximation of the exact martingale certificate. This gives a constructive interpretation of why quadratic certificates are useful, but does not claim that they are necessary.

4 First theoretical results

4.1 A class for which piecewise-linear certificates fail

Theorem (one-sided failure). Let (1) be stopped at the boundary of $[a, b]$, with $g(x)^2 > 0$ in (a, b). Suppose that positive drift occurs arbitrarily close to the left boundary:

$$\text{for every } \delta > 0 \text{ there is } x \in (a, a + \delta) \text{ such that } f(x) > 0. \tag{14}$$

There is no continuous, piecewise-linear certificate V that simultaneously satisfies $V(a) < V(b)$, the kink condition (7), and the supermartingale condition on all interior linear regions.

Proof. The kink condition makes the slopes of V nonincreasing from left to right. If the slope on the first linear region were nonpositive, every later slope would also be nonpositive, contradicting $V(a) < V(b)$. Hence the first slope p is positive. By (14), that region contains a point x^* with $f(x^*) > 0$. At this smooth point,

$$\mathcal{L}V(x^*) = pf(x^*) > 0, \tag{15}$$

contradicting the necessary interior condition. □

There is a symmetric result for a decreasing boundary requirement and negative drift near the right boundary. The theorem also clarifies that failure is not universal: piecewise-linear certificates can work when their slopes have the opposite sign to the drift. For example, for the deterministic Ornstein–Uhlenbeck dynamics $dX_t = -X_t\,dt$, $V(x) = |x|$ has $\mathcal{L}V = -|x| \le 0$ away from its kink.

4.2 Sufficiency for piecewise-quadratic certificates

Let V be continuous and piecewise quadratic. If

1. $\mathcal{L}V \le 0$ in the interior of every quadratic piece,
2. $V'_+(x_i) - V'_-(x_i) \le 0$ at every breakpoint, and
3. $\int_0^t V'(X_s)g(X_s)\,dW_s$ is a martingale,

then (5) immediately shows that $V(X_t)$ is a supermartingale. This is a sufficient condition for a proposed certificate, not a claim that a piecewise-quadratic certificate exists for every SDE or that quadratic degree is necessary.

An unrestricted deep network with quadratic activations would produce high-degree polynomial pieces and make verification expensive. A more promising architecture is therefore a ReLU network whose final layer is smooth. Most of the feature extraction would remain piecewise linear and cheap, while the certificate output would possess the curvature needed in examples such as (9).

5 Verifier scaffold

I implemented a verifier for ReLU Neural Network certificates for one-dimensional SDEs. It currently:

1. directly extracts the "pieces" of a ReLU neural network;
2. checks $\inf_{X_u} V \ge \beta$ and $\sup_{X_0} V \le \alpha$ over unions of intervals;
3. restricts verification to the sub-β basin outside the target;
4. checks $V'f \le -\varepsilon$ on every relevant smooth segment; and
5. checks the concativity Tanaka condition at every relevant kink.

I currently also assume that $f(x)$ is a linear function. I use this assumption because I need to check $\sup_{x\in[l,u]} \mathcal{L}V(x) = \sup_{x\in[l,u]} a_i f(x)$. Perhaps the user would also supply a method `OneDimensionalSDE.drift_bounds(lower, upper) -> (lower, upper)` t hat returns the bounds of $f(x)$ on the interval [`lower`, `upper`].